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Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
interface DeliveryCharge {
    double calculateDeliveryCharge(double foodCost);
}
interface CouponDiscount {
    double calculateDiscount(double foodCost, String couponCode);
}
class FoodOrder implements DeliveryCharge, CouponDiscount{
    @Override
    public double calculateDeliveryCharge(double foodCost){
        if(foodCost >= 500){
            return 0;
        }else{
            return 50;
        }
    }
    @Override
    public double calculateDiscount(double foodCost, String CouponCode){
        if(CouponCode.equals("SAVE10")){
            return foodCost * 0.10;
        }else{
            return 0;
        }
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        double foodCost = sc.nextDouble();
        sc.nextLine();
        String couponCode = sc.nextLine();
        FoodOrder order = new FoodOrder();
        double deliveryCharge = order.calculateDeliveryCharge(foodCost);
        double discount = order.calculateDiscount(foodCost, couponCode);
        double finalBill = foodCost + deliveryCharge - discount;
        System.out.println("Food Cost : " + foodCost);
        System.out.println("Delivery Charge : " + deliveryCharge);
        System.out.println("Discount : " + discount);
        System.out.println("Final Bill : " + finalBill);
        sc.close();
    }
}
```

INPUT

600
SAVE10

OUTPUT

FoodCost : 600.0
DeliveryCharge : 0.0
Discount : 60.0
FinalBill : 540.0

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Room
{
    int roomNumber;
    String guestName;
    double roomRent;
    Room(int roomNumber, String guestName, double roomRent)
    {
        this.roomNumber = roomNumber;
        this.guestName = guestName;
        this.roomRent = roomRent;
    }

    interface ServiceCharge
    {
        double calculateServiceCharge();
    }

    static class HotelBill extends Room implements ServiceCharge
    {
        HotelBill(int roomNumber, String guestName, double roomRent)
        {
            super(roomNumber, guestName, roomRent);
        }
        public double calculateServiceCharge()
        {
            if(roomRent >= 5000)
                return roomRent * 0.12;
            else
                return roomRent * 0.08;
        }
    }

    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int roomNumber = sc.nextInt();
        sc.nextLine();
        String guestName = sc.nextLine();
        double roomRent = sc.nextDouble();
        HotelBill bill = new HotelBill(roomNumber, guestName, roomRent);
        double serviceCharge = bill.calculateServiceCharge();
        double finalBill = roomRent + serviceCharge;
        System.out.println("Room Number : " + roomNumber);
        System.out.println("Guest Name : " + guestName);
        System.out.println("Room Rent : " + roomRent);
        System.out.println("Service Charge : " + serviceCharge);
        System.out.println("Final Bill : " + finalBill);
        sc.close();
    }
}
```

INPUT

```
205
ananya
6000
```

OUTPUT

```
Room Number : 205
Guest Name : ananya
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0
```